C4 model · Level 2 of 4

Vesta: Level 2: Containers

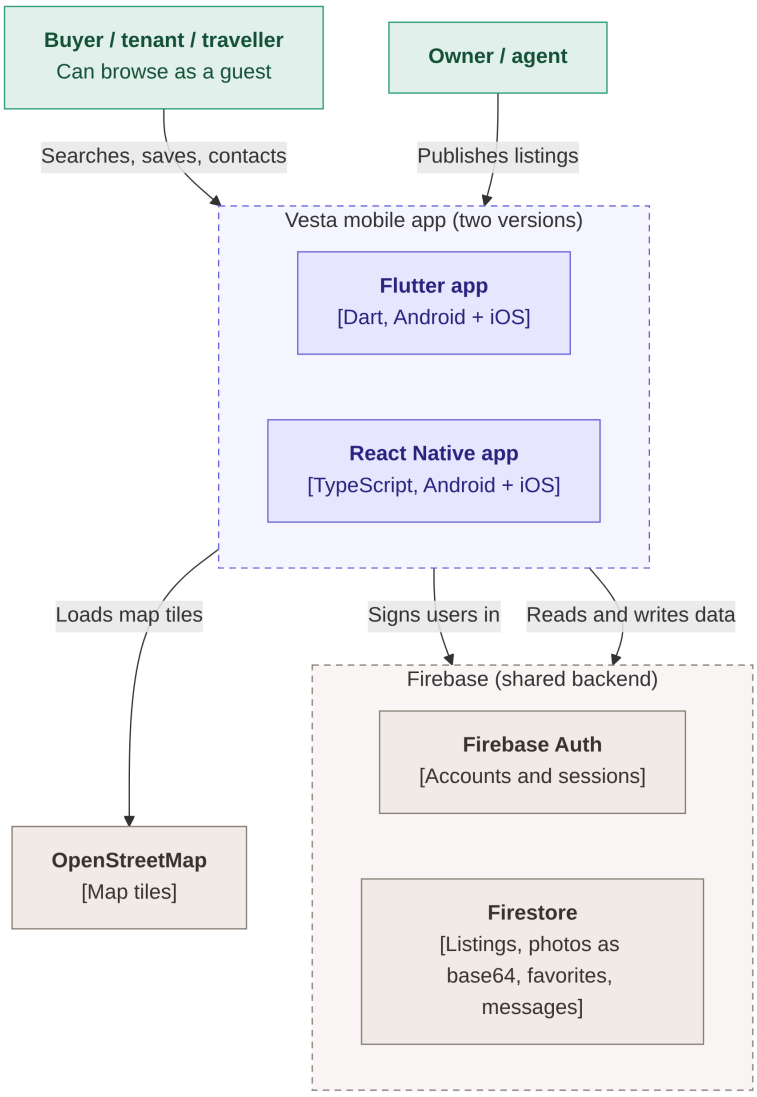
Project: real estate mobile app (Flutter and React Native)

Question answered by this level: What are the main parts of Vesta, and how do they communicate?

Description

This level zooms inside Vesta, like going from the country to the city. A container is a part that runs on its own: an app, a database, a service. It has nothing to do with Docker.

Diagram



Elements of the diagram

Element	Role
Flutter app	First version, written in Dart, for Android and iOS. This is the one published on Google Play.
React Native app	Second version, written in TypeScript, for Android and iOS (academic demo).
Firebase Auth	Handles sign-up, sign-in and sign-out.
Firestore	Database: listings, photos (base64, max 5 per listing), favorites, conversations and messages.
OpenStreetMap	Provides the map tiles used by both apps. The credit "© OpenStreetMap contributors" is shown.

Key points

- Both apps use the same Firebase: the backend is built only once.
- A user uses one version or the other, never both at the same time.
- No Firebase Storage (paid plan): photos are in Firestore. No server push in the MVP: message alerts are shown in the app.
- Architecture rule inside each app: Screen -> Controller -> Repository -> Firebase.